

Variational Autoencoders

$$\max_{\theta} \log p_{\theta}(x) \xrightarrow{\text{②}} \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z))$$

$$\begin{aligned} \log p_{\theta}(x) &= \log p_{\theta}(x) \cdot \int \underline{q_{\phi}(z|x)} dz \\ &= \int \underline{q_{\phi}(z|x)} \cdot \log p_{\theta}(x) \underline{dz} = \mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x)] \\ &= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{p_{\theta}(z|x)} \right] \end{aligned}$$

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$$\max_{\theta} \log p_{\theta}(x) \rightarrow \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) || p(z))$$

$$\begin{aligned} \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{p_{\theta}(z|x)} \right] &= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \left\{ \frac{p_{\theta}(x, z)}{p_{\theta}(z|x)} \underbrace{\frac{q_{\phi}(z|x)}{q_{\phi}(z|x)}}_{=1} \right\} \right] \\ &= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} \right] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right] \end{aligned}$$

$$\overset{\text{max}}{\log p_{\theta}(x)} = (\text{ELBO}) + \underbrace{D_{\text{KL}}(q_{\phi}(z|x) || p_{\theta}(z|x))}_{\geq 0} \quad \therefore \log p_{\theta}(x) \geq \text{ELBO}$$

$$\begin{aligned} \text{ELBO} = \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} \right] &= \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p_{\theta}(x|z) \cdot p(z)}{q_{\phi}(z|x)} \right] = \mathbb{E}_{q_{\phi}(z|x)} \left[\log p_{\theta}(x|z) \right] - \underbrace{\mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{q_{\phi}(z|x)}{p(z)} \right]}_{D_{\text{KL}}(q_{\phi}(z|x) || p(z))} \end{aligned}$$